

PRODUCER CONSUMER PROBLEM: (USING SEMAPHORES)

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#define BUFFER_SIZE 5
int buffer[BUFFER_SIZE];
int in = 0, out = 0;
sem_t mutex, empty, full;
void* producer(void* arg) {
    int item;
    for (int i = 0; i < 10; i++) {
        item = i;
        sem_wait(&empty);
        sem_wait(&mutex);
        buffer[in] = item;
        printf("Produced: %d\n", item);
        in = (in + 1) % BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&full);
        sleep(1);
    }
    return NULL;
}
void* consumer(void* arg) {
    int item;
    for (int i = 0; i < 10; i++) {
        sem_wait(&full);
        sem_wait(&mutex);
        item = buffer[out];
        printf("Consumed: %d\n", item);
        out = (out + 1) % BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&empty);
        sleep(2);
    }
    return NULL;
}
```

```
int main() {  
    pthread_t p, c;  
    sem_init(&mutex, 0, 1);  
    sem_init(&empty, 0, BUFFER_SIZE);  
    sem_init(&full, 0, 0);  
    pthread_create(&p, NULL, producer, NULL);  
    pthread_create(&c, NULL, consumer, NULL);  
    pthread_join(p, NULL);  
    pthread_join(c, NULL);  
    sem_destroy(&mutex);  
    sem_destroy(&empty);  
    sem_destroy(&full);  
    return 0;  
}
```